



Sensor-free Microclimate Monitoring Using Existing LoRaWAN Signal Characteristics

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ABSTRACT

Long-range wireless protocols enable large-scale Internet of Things (IoT) applications and vast sensor deployments. The growth of such deployments presents a unique opportunity for wireless sensing, as these devices generate spatially distributed wireless signals. Instead of today's convention of new projects requiring new hardware, expensive deployments, and increased ongoing maintenance costs, we propose a technique to leverage existing wireless signals from already deployed sensors to perform wide-area sensing—for free. And unlike typical wireless sensing approaches that generate new wireless signals for sensing, our approach uses unmodified devices and wireless signals already captured by LoRaWAN gateways to monitor the environment.

We demonstrate the practicality of this technique by designing and building a microclimate monitoring system that targets temperature using an XGBoost regression model trained on received signal strength indicator (RSSI) and signal-to-noise ratio (SNR). Through eight months of data collection in a four-season humid subtropical city, we reveal a dynamic correlation between temperature and LoRaWAN signal characteristics that vary with seasonal climate changes. Our methodology estimates local temperature with a mean absolute error (MAE) of 0.99 °C, compared to 1.41 °C from the nearest city weather station (CWS). Our system particularly provides more accurate estimates than the CWS during periods of peak summer heat and deep winter cold extremes. These results highlight the feasibility, efficiency, and scalability of leveraging existing LoRaWAN networks as a sustainable, sensor-free solution for urban environmental monitoring.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems, Sensor networks.**

KEYWORDS

RF Sensing, LoRaWAN, RSSI, SNR, Microclimate Monitoring

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1 INTRODUCTION

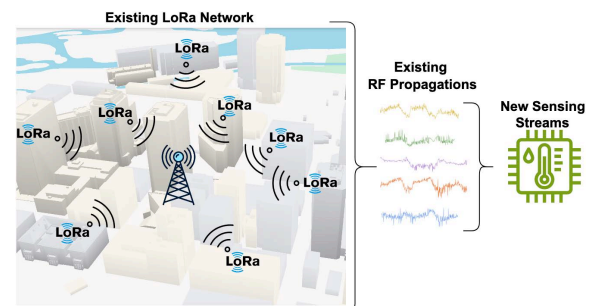


Figure 1: Our sensing system leverages existing RF propagation from already deployed LoRaWAN infrastructure to extract new sensing streams without the need for additional sensor deployments. This sensor-free approach is compatible with any IoT device connected to the gateway, requiring no modifications to the existing hardware or protocols.

The ubiquity of wireless communication networks, alongside the widespread adoption of low-cost IoT devices, presents a unique opportunity to rethink traditional sensing paradigms. On the other hand, deploying dedicated sensor networks at scale, such as across entire cities, remains prohibitively expensive due to high installation costs, ongoing maintenance, and logistical challenges in managing large, distributed sensor systems. Prior research has shown that environmental factors, such as presence of obstacles or variations in temperature, fog, or rain can influence radio frequency (RF) signal propagation [5, 7], raising the question of whether these



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naturally occurring signal variations could be leveraged for sensing without the need for additional sensor deployments. Building on this insight, we investigate the feasibility of repurposing existing RF infrastructure to extract environmental information from signals already in use, enabling new sensing capabilities without modifying the underlying communication systems. This approach offers a more scalable and cost-effective alternative for many environmental monitoring applications.

Researchers have recently used RF signals for diverse wireless sensing applications [3, 11, 15]. However, most of these systems rely on custom signal generation or specialized hardware deployment. In contrast, repurposing signals originally intended for communication offers an opportunity to extract new sensing streams, enabling environmental inference without additional transmissions or infrastructure changes.

Among existing wireless infrastructures, Long Range (LoRa) Wide Area Network (LoRaWAN) presents a compelling opportunity for repurposing existing signals. The rapid growth of IoT has led to widespread deployment of low-power wide-area networks (LPWANs), particularly LoRaWAN, to support smart applications [9, 19, 21, 24]. LoRaWAN offers long-range communication, low power consumption, and strong signal penetration, making it ideal for wide-area coverage with minimal energy usage. With an estimated 3.5 billion LPWAN connections by 2030 [2], the increasing density of LoRa-enabled devices provides a strong foundation for repurposing existing signals for environmental sensing. As LoRa transmissions propagate across long distances and varying environmental conditions, signal characteristics such as RSSI and SNR can reflect ambient influences, offering a rich, untapped data source for sensing (see Figure 1).

Repurposing LoRaWAN signals for sensing can unlock new large-scale applications, particularly in urban microclimate monitoring. High-resolution temperature data is essential for capturing fine-grained urban dynamics, including heat islands, local weather variability, green space management, smart irrigation, and climate resilience planning [6, 12]. Traditional sensing approaches often rely on dense networks of physical sensors, which are costly to deploy and maintain. In contrast, we leverage the growing presence of LoRaWAN infrastructure to passively measure local temperature by analyzing how environmental conditions affect LoRa signal properties. This enables real-time sensor-free micro-temperature estimation using existing communication networks.

A key challenge in repurposing LoRa signals for temperature sensing is the dynamic behavior of RSSI and SNR under changing environmental conditions. Our long-term data collection reveals that temperature fluctuations affect signal strength non-linearly, with clear seasonal transitions (see Section 3). Another challenge lies in gateway placement: gateways, which receive transmissions from end-nodes and

forward them to the network, exhibit different signal stability depending on whether they are deployed indoors or outdoors. Physical obstructions and interference lead to signal variability that complicates modeling. To address these challenges, we develop an XGBoost regression model trained on long-term LoRa signal metadata to infer ambient temperature. Our fine-tuned model achieves an MAE of 0.99 °C, compared to 1.41 °C from the city’s official weather station (CWS) over the same period against local ground-truth. It performs particularly well during temperature peaks in summer and dips in winter, periods that are critical for microclimate applications, including heat island, smart irrigation, and climate-responsive planning.

Unlike many conventional wireless sensing systems, our system does not require new sensor deployments, any changes to network configurations, or altering communication protocols. Instead, it leverages natural RF signal fluctuations caused by environmental changes, transforming existing infrastructure into a real-time, sensor-free environmental sensing platform. This method reduces cost, energy use, and eliminates maintenance overhead, while enhancing the utility of existing LoRaWAN deployments. Our findings demonstrate not only the feasibility of precise sensor-free microclimate monitoring, but also the broader potential of LoRaWAN networks for scalable, low-cost environmental monitoring.

2 RELATED WORK

Microclimate Monitoring Approaches. Traditional microclimate monitoring typically involves deployment of dedicated environmental sensors, incurring high installation and maintenance costs [12]. Recent IoT-based systems aim to mitigate these burdens using low-cost, distributed sensing nodes [16], yet they still rely on deploying physical hardware. In contrast, our work eliminates additional sensor deployment by repurposing existing LoRaWAN infrastructure.

Wireless Sensing using Existing Networks. Recent work has demonstrated the potential of repurposing existing wireless communication infrastructures for sensing tasks without modifying communication protocols. Examples include occupancy detection using Wi-Fi [17, 23] and urban traffic monitoring leveraging LoRaWAN RSSI patterns [3]. These studies highlight the broader opportunity to extract environmental insights from existing communication signals.

Environmental Effects on LoRa Network Performance. LoRaWAN signal characteristics (RSSI and SNR) are known to be sensitive to environmental conditions such as temperature, humidity, fog, and precipitation [5, 7, 18, 20]. While most studies have focused on communication reliability, a few explored using these variations for sensing environmental conditions Bhat et al. [4] used deep learning to estimate ambient conditions from LoRaWAN data, but their limited dataset (100 hours) restricts generalizability. In contrast, our

work builds on this direction using long-term signal data and demonstrates that RSSI and SNR trends can reflect seasonal environmental patterns, enabling accurate, sensor-free microclimate monitoring.

3 SYSTEM DESIGN

LoRaWAN is a low-power, wide-area communication protocol widely adopted for scalable IoT applications. It operates in ISM radio bands using chirp spread spectrum (CSS) modulation, which supports long-range and low-power transmissions [2]. It extends LoRa by enabling secure, bi-directional communication between IoT devices and gateways through adaptive data rate (ADR), optimizing parameters such as spreading factor and transmission power [10]. While LoRaWAN was originally designed for communication, LoRaWAN signal characteristics such as RSSI and SNR are sensitive to environmental changes.

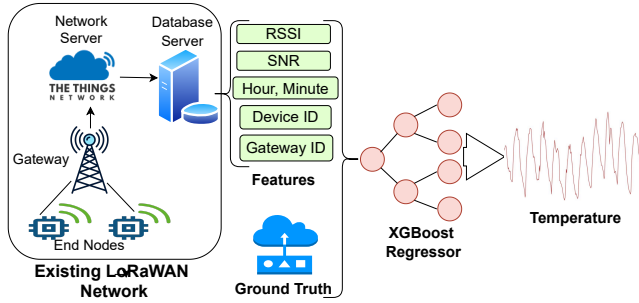


Figure 2: System overview.

Our system leverages the influence of environmental conditions on LoRaWAN’s signal characteristics to enable microclimate monitoring. Instead of deploying additional environmental sensors, we repurpose existing LoRaWAN RSSI, SNR, and time information, reported by the gateways to The Things Network (TTN) server. We use these passively collected features as inputs to a trained supervised regression model to estimate the local temperature. By learning the relationship between LoRa signal characteristics and ground-truth measurements from a weather station, our system enables real-time temperature measurement across the existing LoRaWAN network. Figure 2 illustrates the components of our system architecture.

The system consists of four key components, as shown in Figure 2: (1) the existing LoRaWAN network deployed in the area, (2) TTN server, (3) a data preprocessing pipeline that extracts features from RSSI and SNR, and (4) a supervised regression model for temperature estimation. Sensors already deployed in the environment periodically transmit encrypted data packets to nearby LoRaWAN gateways, which forward them to the TTN server. Although the application data is not accessible due to encryption, the gateways report transmission characteristics such as RSSI and SNR for each received packet. We combine the RSSI and SNR measurements with

time information and then extract features to feed to the regression model. The model estimates the ambient temperature which we evaluate by comparing predictions against measurements from our locally deployed weather stations.

Data Collection and Preprocessing. We collect time-stamped RSSI and SNR values for each device-gateway pair and use them to construct the feature space. To preserve the original granularity, we merge all device-gateway pair data into a single dataset longitudinally, resulting in finer temporal resolution. Since each device transmits once every five minutes, we obtain one sample per pair, yielding n samples per five-minute window, where n is the total number of pairs (e.g., 2 gateways and 9 devices).

Each sample includes raw RSSI and SNR values, the corresponding timestamp, and the encoded identifiers of the device-gateway pair. To capture temporal patterns, we decompose the timestamp into separate hour and minute features. Then, we exclude any device-gateway pair whose data has more than 50% missing packets to ensure reliability. We sort the remaining data by timestamp and normalize it over sliding windows of different sizes using z-score normalization. Since the data exhibits daily temporal patterns, we apply sliding window normalization over 24-hour, 12-hour, 6-hour, 3-hour, and 1-hour windows to preserve seasonality and allow selection of the most informative normalized signals as features. Lastly, we label each sample with the corresponding temperature measured by the AcuRite weather station [1] located at the center of the deployment area.

Modeling Approach. To estimate temperature from LoRa signal metadata, we evaluate multiple regression algorithms, including neural networks, linear regression, a Markov-based regressor, random forest, and XGBoost regressor. Among evaluated algorithms, the XGBoost regressor ultimately provided the best combination of accuracy, robustness, and generalizability. Its robustness to outliers and ability to model non-linear relationships make it well-suited for this application. We train and evaluate the models using supervised learning with the ground-truth labels provided by the weather station. Our final model is trained on 80% of the eight-month dataset and evaluated against 20% held-out windows (10% evaluation and 10% test). To robustly evaluate model performance across varied environmental conditions, we adopted a temporally stratified training and validation approach. Additionally, we incorporate historical weather station data in a fine-tuning step to improve prediction accuracy during periods of minimal RSSI variation, a phenomenon we refer to as the “floor effect”, which we discuss later in this section.

Design Challenges. During a four-month pilot study (September 2023 to January 2024), we observed that LoRa signal behavior—specifically RSSI and SNR—varies not only with changing ambient conditions, but also significantly across different deployment environments. We summarize the key

findings from this pilot study that helped us understand how temperature influences signal characteristics as follows.

(1) *Effects of Environmental Setup on LoRa Signal Characteristics*: Our pilot study observations demonstrate that the physical environment surrounding a gateway affects LoRa signal propagation. Factors such as the gateway's physical location, nearby structures, and natural obstructions (e.g., trees or hills) alter the propagation path and signal strength. For instance, the outdoor gateway, mounted on the roof of a five-story building, exhibits more stable and higher average RSSI and SNR due to unobstructed line-of-sight conditions. In contrast, the indoor gateway, installed on the second floor of an office building, experiences greater signal attenuation from walls and other obstructions. On average, received RSSI values at the indoor gateway are 6 dBm lower than those at the outdoor one, and the indoor gateway's SNR distribution spans a wider range, indicating greater fluctuation in signal quality. Additionally, weather conditions, such as rain and extreme temperatures, can affect signal behavior differently, depending on whether the gateway is indoors or outdoors. These variations in signal characteristics across different deployment environments present a unique challenge in developing a robust model that can adapt to and accurately interpret signal data from both indoor and outdoor deployments.

(2) *Dynamic Weather Impacts on RSSI Behavior*: We analyze environmental factors recorded by the weather station, including temperature, humidity, precipitation, barometric pressure, and wind speed, to understand their influence on LoRa signal behavior. Among these, temperature consistently shows the strongest correlation with RSSI variation.

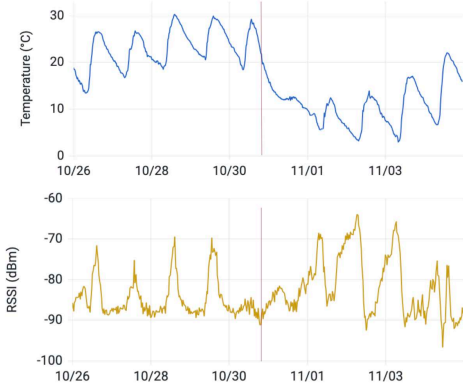


Figure 3: Example of a detected temperature change point and its impact on RSSI behavior.

As illustrated in Figure 3, a sudden temperature drop on October 30th causes a clear shift in RSSI patterns. Before the change point, RSSI peaks align with afternoon temperature highs around 14:00; after the drop, the peak shifts to early morning around 7:00, reversing the correlation.

We identify seven such change points during our four-month pilot study using a dynamic programming-based segmentation approach. These shifts highlight how temperature not only affects RSSI but also alters its temporal patterns, making it a critical feature for sensing. However, this variability also introduces modeling challenges that require a system that can adapt to seasonal transitions and shifting signal dynamics for reliable microclimate monitoring.

(3) *RSSI floor effect*: In our pilot study, we also observe that during moderate weather conditions, RSSI tends to stabilize around a specific value, for example, approximately -90 dBm for the outdoor gateway, as shown in Figure 3. We refer to this phenomenon as the “floor effect”, where the lack of signal variation reduces the model’s ability to distinguish temperature changes, thus affecting the performance of the system during periods of mild temperature.

In summary, addressing these challenges requires careful preprocessing and robust feature engineering to ensure the model can learn generalizable patterns across heterogeneous gateway conditions and the dynamic influence of ambient temperature on LoRa signal behavior.

4 METHODOLOGY

To demonstrate our sensor-free microclimate monitoring approach, we deploy a LoRaWAN network across a 380×400 meter area on a university campus in a four-season humid subtropical city. The deployment includes two LoRa gateways (one indoor, one outdoor), nine off-the-shelf LoRa-enabled sensors as end-nodes, and a centrally located weather station for ground-truth measurements (Figure 4).

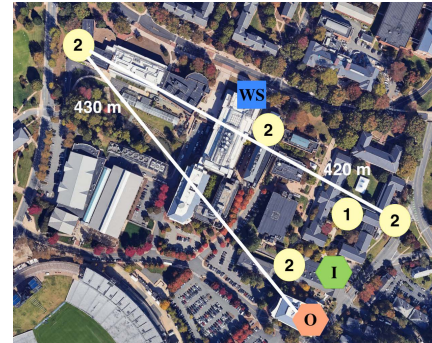


Figure 4: LoRaWAN deployment map showing two gateways represented by orange (outdoor (O)) and green (indoor (I)) hexagons, nine end-nodes indicated by yellow circles (each circle represents the number of deployed end-nodes in that area), and a weather station (WS) for ground truth measurements depicted as a blue square.

We position one Cisco IXM-LPWA-900-16-K9 [8] gateway outdoors on the roof of a five-story office building to maximize coverage, and another indoor on the second floor of a building beside a window to ensure the system performance

in an obstructed environments. As end-nodes, we use Lino-vision S500TH [14] and S500CO₂ [13] sensors. We place four pairs of sensors in four distinct buildings, each consisting of one sensor indoors and one outdoors. An additional sensor is installed in an open corridor adjacent to a yard.

All sensors transmit data every five minutes using default communication settings. The network operates in the 902–928 MHz ISM band with ADR enabled, and each sensor transmits at a nominal power of 20 dBm. The Cisco gateways use the Common Packet Forwarder and are configured to communicate with TTN server [21], forwarding LoRaWAN uplink messages and receiving LoRaWAN downlink messages. This setup reflects a realistic, community-compatible LoRaWAN deployment.

Unlike software-defined radios such as USRPs, which allow full control over radio parameters, our setup is constrained to the data available through commercial off-the-shelf gateways and sensors. We do not have access to raw IQ samples or physical layer data. Instead, we rely solely on metadata that gateways provide through TTN, including RSSI and SNR. These constraints mirror the typical limitations of real-world deployments and highlight the feasibility of our sensor-free approach under realistic conditions.

We collect RSSI and SNR measurements for each packet received by the gateways and align them with temperature data reported by an AcuRite Iris weather station [1]. The weather station is installed at the center of the deployment area and reports data at five-minute intervals, allowing for accurate synchronization with the LoRa signal data.

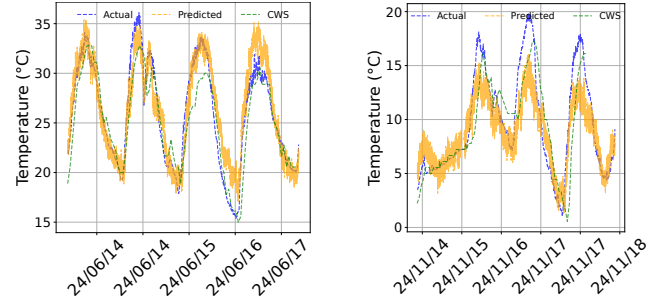
We conduct data collection continuously over an eight-month period, from June 10, 2024 to February 14, 2025. This duration allows us to analyze the influence of seasonal, structural, and spatial variability on LoRa signal characteristics across diverse environmental conditions.

5 EVALUATION

In this section, we present a comprehensive and long-term experiment we conduct to validate our system in different ambient conditions.

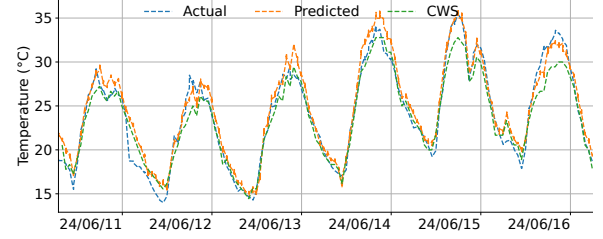
Model Training. We evaluate our sensor-free microclimate sensing system by developing two XGBoost regression models (warm season and cold season models) trained solely on RSSI, SNR, and their associated time-of-day (minute and hour) to retain daily seasonal patterns. We fine-tune the XGBoost models' hyperparameters using grid search to MAE.

System Performance. To benchmark our system, we compare models predictions against those from the CWS, located 13.5 km away at the airport. We achieve MAE of 2.00 °C over the warm period model and 2.76 °C over the cold period model when compared to our ground truth from the locally deployed AcuRite weather station [1]. As shown in Figure 5, our system provides more accurate local estimates than the

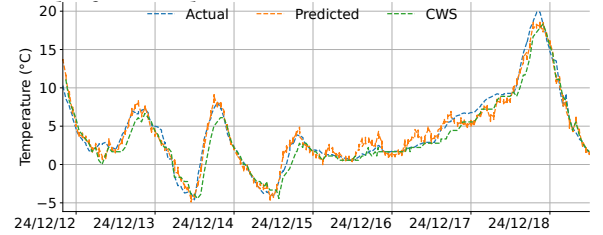


(a) Warm weather prediction. (b) Cold weather prediction.

Figure 5: Examples of warm and cold models' predictions only using LoRa signal metadata compared to the ground-truth and the CWS, indicating the effectiveness of our sensor-free approach during periods of extreme weather.



(a) Our system provides more accurate estimates of local temperature on June 15 and 16, particularly during



(b) Our system achieves higher accuracy during the coldest days in the test dataset (December 12–14).

Figure 6: Temperature prediction comparison between our system and the CWS for two test segments: (a) summer heat and (b) winter cold. Local ground-truth measurements (actual) are also shown for reference.

CWS during critical periods, particularly during temperature peaks in warm seasons and dips in cold seasons. We also observe intervals where it is raining near the CWS but dry and sunny in our deployment area, leading to temperature differences significantly higher than the MAE observed between our system and the local weather station. These high-variance periods are especially relevant for microclimate applications such as outdoor planning, smart irrigation, and heat island detection.

These findings highlight the value of localized, sensor-free monitoring with no additional deployment cost, over traditional distant, centralized weather stations.

Fine-Tuned microclimate model. While our system performs well in high-variance periods, it does not perform as well during moderate periods due to the “floor effect” (see Figure 3). During these periods, the lack of signal variation limits the model’s ability to accurately infer temperature. To address this limitation, we fine-tune a single model using historical hourly data from the CWS, obtained via Weather Underground [22]. This auxiliary signal acts as a complementary correction factor, enhancing prediction accuracy during flat-RSSI periods. The final configuration of our fine-tuned model uses a maximum tree depth of 6, a minimum leaf sample size of 20, a learning rate of 0.1, and the `reg:squarederror` objective function. To ensure robustness across diverse temperature conditions, we partition the dataset into four equal temporal segments. Within each segment, we allocate the first 80% of samples for training, the next 10% for validation, and the remaining 10% for testing. This stratified split ensures that each phase of model development—training, tuning, and evaluation—encounters varying environmental conditions, including transitions across cold and warm seasons.

After fine-tuning, our system achieves an overall MAE of 0.99 °C, compared to 1.41 °C from the CWS, and demonstrates improved accuracy across both extreme and mild conditions. To ensure robustness, we evaluate the model over four temporally distinct test segments that span a wide temperature range from -4.4 °C to 37.8 °C, capturing the full seasonal and daily variation in our deployment region (see Figure 6). These results demonstrate the accuracy and generalization capability of our fine-tuned model, which performs reliably across all temperature thresholds and effectively addresses the key design challenges outlined in Section 3, making it well-suited for real-world microclimate monitoring.

6 CONCLUSION

This study demonstrates the innovative potential of utilizing LoRaWAN’s signal characteristics for environmental monitoring, revealing that existing RF communication infrastructure can serve as a powerful, sensor-free tool for extracting environmental information. By repurposing existing LoRaWAN metadata, specifically RSSI and SNR, our approach transforms the infrastructure originally designed for communication into a passive, cost-effective, and energy-efficient environmental sensing platform. While we focus on micro-temperature estimation, this methodology opens the door to a broader class of applications, including humidity and precipitation.

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¹www.cyberinitiative.org

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